

SOFT COMPUTING

ASSIGNMENT -8

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1.

a). Write a Python program to train a Back Propagation Neural Network (BPNN) for classifying handwritten digits.

```
import keras
keras.datasets.mnist.load_data(path="mnist.npz")
from keras.datasets import mnist
from keras.models import Sequential
from keras.layers import Dense, Flatten
from keras.utils import to_categorical
from keras.optimizers import Adam

(training_images, training_labels), (testing_images, testing_labels)
= mnist.load_data()

training_images = training_images.reshape((training_images.shape[0],
28 * 28)).astype('float32') / 255
testing_images = testing_images.reshape((testing_images.shape[0], 28
* 28)).astype('float32') / 255

training_labels = to_categorical(training_labels)
testing_labels = to_categorical(testing_labels)

model = Sequential()
model.add(Dense(128, activation='relu', input_shape=(28 * 28,)))
model.add(Dense(64, activation='relu'))
model.add(Dense(10, activation='softmax'))

model.compile(optimizer=Adam(), loss='categorical_crossentropy',
metrics=['accuracy'])
```

```
model.fit(training_images, training_labels, epochs=10, batch_size=60,
verbose=2)

test_loss, test_acc = model.evaluate(testing_images, testing_labels,
verbose=0)

print(f'Accuracy of BPPN Model Upon Testing is : {test_acc *
100:.2f}%')
```

```
Epoch 1/10
1000/1000 - 2s - 2ms/step - accuracy: 0.9206 - loss: 0.2764
Epoch 2/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9672 - loss: 0.1091
Epoch 3/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9771 - loss: 0.0746
Epoch 4/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9817 - loss: 0.0574
Epoch 5/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9857 - loss: 0.0448
Epoch 6/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9885 - loss: 0.0352
Epoch 7/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9905 - loss: 0.0282
Epoch 8/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9913 - loss: 0.0266
Epoch 9/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9930 - loss: 0.0212
Epoch 10/10
1000/1000 - 1s - 1ms/step - accuracy: 0.9945 - loss: 0.0161
Accuracy of BPPN Model Upon Testing is : 97.44%
```

b). Write a Python program to train a Multi Layer Perceptron (MLP) for classifying handwritten digits. Compare the results of both neural networks.

```
from keras.datasets import mnist
from keras.models import Sequential
from keras.layers import Dense
from keras.utils import to_categorical
from keras.optimizers import Adam

(training_images, training_labels), (testing_images, testing_labels)
= mnist.load_data()

training_images = training_images.reshape((training_images.shape[0],
28 * 28)).astype('float32') / 255
testing_images = testing_images.reshape((testing_images.shape[0], 28
* 28)).astype('float32') / 255

training_labels = to_categorical(training_labels)
testing_labels = to_categorical(testing_labels)

model_mlp = Sequential()
model_mlp.add(Dense(512, activation='relu', input_shape=(28 * 28,)))
model_mlp.add(Dense(256, activation='relu'))
model_mlp.add(Dense(10, activation='softmax'))

model_mlp.compile(optimizer=Adam(), loss='categorical_crossentropy',
metrics=['accuracy'])

model_mlp.fit(training_images, training_labels, epochs=10,
batch_size=60, verbose=2)

test_loss_mlp, test_acc_mlp = model_mlp.evaluate(testing_images,
testing_labels, verbose=0)
```

```
print(f'Accuracy of MLP Model Upon Testing is : {test_acc_mlp *  
100:.2f}%')
```

Epoch 1/10

1000/1000 - 4s - 4ms/step - accuracy: 0.9425 - loss: 0.1928

Epoch 2/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9759 - loss: 0.0782

Epoch 3/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9839 - loss: 0.0505

Epoch 4/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9872 - loss: 0.0391

Epoch 5/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9901 - loss: 0.0300

Epoch 6/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9921 - loss: 0.0240

Epoch 7/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9925 - loss: 0.0224

Epoch 8/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9944 - loss: 0.0171

Epoch 9/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9942 - loss: 0.0173

Epoch 10/10

1000/1000 - 3s - 3ms/step - accuracy: 0.9955 - loss: 0.0135

Accuracy of MLP Model Upon Testing is : 98.04%